



Polystyrene microplastic exposure modulates gut microbiota and gut-liver axis in gilthead seabream (*Sparus aurata*)

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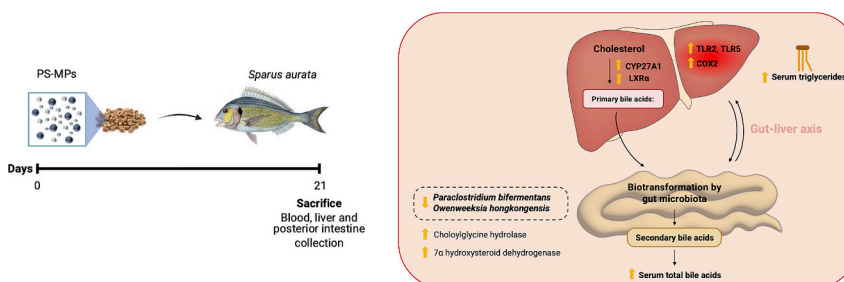
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HIGHLIGHTS

- MPs alter gut microbiota architecture and related predicted functions.
- MPs impact on bile acid metabolism and synthesis.
- MPs stimulate hepatic innate immunity.
- Gut-liver axis is involved in MP-induced toxicity in gilthead seabream.

GRAPHICAL ABSTRACT



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ABSTRACT

Microplastics (MPs) are a threat of growing concern for living organisms as they exist in all ecosystems. The bidirectional communication between the gut, its microbiota, and the liver, has been conceptualized as gut-liver axis and may be influenced by environmental factors. MPs can cause intestinal and hepatic injuries, but there is still limited research exploring their impact on gut-liver axis. The aim of this study was to assess the effects of MP ingestion on gut-liver axis balance in gilthead seabream (*Sparus aurata*) fed with a diet enriched with polystyrene (PS)-MPs (0, 25, or 250 mg/kg b.w./day) for 21 days. PS-MPs affected the composition of gut microbiota, enhancing the evenness of gut microbial species. We also observed the impoverishment of core microbiota, suggesting reduced stability and permanence of microbiota members. Furthermore, PS-MPs reduced

Abbreviations: ANOSIM, Analysis of similarities; ASVs, Amplicon Sequence Variants; BAs, bile acids; BLAST, Basic Local Alignment Search Tool; CON, control; COX, Cyclooxygenase; CYP, Cytochrome; DAMPs, danger-associated molecular patterns; KEGG, Kyoto Encyclopedia of Genes and Genomes; LXRs, liver X nuclear receptors; MPs, microplastics; MyD88, Myeloid differentiation factor 88; PCoA, Principal Component Analysis; PICRUSt2, phylogenetic investigation of communities by reconstruction of unobserved states; PS, polystyrene; QIIME2, Quantitative Insights Into Microbial Ecology; RAS, Recirculating Aquaculture System; SILVA DB, SILVA database; TBA, total bile acids; TGL, triglycerides; TLRs, toll-like receptors.

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predominant bacteria in the gut of gilthead seabreams, increasing low-abundance species, including potential harmful taxa. On the other hand, PS-MPs increased the gene expression of immune and inflammatory mediators (i.e., *TLR2*, *TLR5*, and *COX-2*) in the liver. PS-MP exposure also increased serum triglycerides and bile acids (BAs) without modifying cholesterol. Moreover, the hepatic BA metabolism was impacted by PS-MPs which increased the expression of genes involved in primary BA kinetic (i.e., *CYP27A1* and *LXRα*), which in turn can modulate intestinal microbial community. Indeed, PICRUSt2 mapping of BA-related functions predicted the increase of factors involved in BA metabolism. Specifically, K01442 (choloylglycine hydrolase) and K00076 (7 α -hydroxysteroid dehydrogenase) were augmented by PS-MPs, suggesting a possible adaptation or co-evolution of gut microbiota to the modified hepatic BA metabolism. Thus, the obtained results showed that ingested PS-MPs impact the gut microbiota architecture and functions, the hepatic innate immunity, and the BA metabolism, suggesting the involvement of the gut-liver axis in MP-induced toxicity.

1. Introduction

Microplastics (MPs) are synthetic solid particles or polymer matrices, with regular or irregular shapes, insoluble in water, ranging in size from 100 nm to 5 mm. They can be of either primary or secondary origin (Frias and Nash, 2019). Primary MPs are intentionally produced in small sizes, such as microbeads in personal care products or synthetic fibers released from clothing (Le et al., 2022). Secondary MPs result from the degradation of larger plastic items over time, weathered by sunlight, waves, and mechanical processes (Issac and Kandasubramanian, 2021). Due to their ubiquitous occurrence in the environment, MPs are recognized as emerging pollutants with a high ecotoxicological impact, and their occurrence in the aquatic environment is considered an urgent global issue (Xu et al., 2020; Ahmed et al., 2021).

Fish introduce MPs mainly through ingestion both directly, by surrounding environmental uptake, and indirectly, by feeding on contaminated prey (Hasegawa and Nakaoka, 2021). Then, MPs can accumulate in the intestinal tract (Cocci et al., 2022) but also translocate in other organs, including the liver (McIlwraith et al., 2021). The occurrence and the extent of translocation processes, as well as the pathways involved in MP distribution and transport, are strictly related to the particle size (Ma et al., 2021). The MP presence in the gut and liver results in detrimental effects on tissue function and homeostasis as reported by several studies (e.g., Feng et al., 2021; Hu et al., 2022; Huang et al., 2021; Yin et al., 2022). In the gut, we recently reported the inflammatory effect, the immune alteration, the redox unbalance, and the impairment of intestinal barrier integrity, in gilthead seabreams fed on polystyrene (PS)-MP enriched diet (Del Piano et al., 2023a; Del Piano et al., 2023b). The inflammatory reaction and the alteration of barrier integrity in the gut may facilitate the passage of external substances, danger-associated molecular patterns (DAMPs), and MPs themselves into circulation, reaching the liver and potentially activating hepatic innate immune response through toll-like receptors (TLRs) pathway. In the liver, MPs affect several indicators of oxidative stress and inflammation, inducing hepatocyte necrosis and metabolic alterations (Del Piano et al., 2024; Sun et al., 2024). However, the critical role of the gut-liver axis in MP-induced intestinal and hepatic toxicity has been overlooked in both aquatic and terrestrial models (Yin et al., 2023; Zhao et al., 2023). Gut-liver axis is a bidirectional crosstalk through the portal circulation and bile ducts, that regulates gut and liver health and disorders. The gut and liver mutually shape factors such as nutrients, metabolites, and bile acids (BAs), controlling metabolic and immune system responses, also modulating bacterial communities' architecture and role (Tilg et al., 2022).

Gut microbiota is a fundamental biological system for the regulation of several physiological functions of the organism. It is recognized as a metabolic organ that enables the uptake of nutrients and energy from ingested food and supplies products that control the host's metabolism. Gut microbiota is also able to counteract the effects of several stressors among which environmental pollutants (Evariste et al., 2019). MPs disrupt the organization of intestinal flora in orally exposed mice (Li et al., 2020), chickens (Li et al., 2023), and fish (Santos et al., 2022), impacting bacterial communities' composition and immune response.

The MP-induced microbiota dysbiosis is accompanied by the alteration of metabolic mediators and pathways, such as those related to lipids and BAs (Jin et al., 2019; Wen et al., 2024).

Primary BAs are synthesized in the liver from cholesterol, excreted into bile, and then biotransformed into secondary BAs by gut microbiota. BAs reabsorbed in the gut return to the liver through the portal vein, completing the enterohepatic circulation. Thus, BAs constitute a liver-gut crosstalk regulating hepatic metabolism, shaping gut microbiota structure and function, and maintaining intestinal integrity (Tilg et al., 2022). Notably, BAs shape gut microbiome diversity, inducing the growth of microbes that metabolize BAs and contextually preventing the growth of bile-sensitive microbes. The metabolization of primary BAs into secondary BAs mainly takes place in the posterior tract of the intestine of vertebrates and supplementation with BAs has been showed to specifically modulate the microbiota composition of posterior intestine in gilthead seabream (Hofmann et al., 2010; Ruiz et al., 2023).

To date, there is limited research exploring how intestinal microbiome deconstruction due to MP ingestion can alter gut microbiome activity and induce functional abnormalities, among which liver metabolic alterations.

The aim of this study was to examine the potential relations between alterations in gut bacterial communities and modifications of hepatic functions in the teleost fish gilthead seabream exposed to PS-MPs. We focused on the expression of genes involved in primary BA kinetic, which in turn modulate microbial community and secondary BA production.

2. Materials and methods

2.1. Fish and keeping conditions

Gilthead seabream (*Sparus aurata* Linnaeus, 1758) was selected for the study, since it is well-recognized as a suitable experimental model for toxicological investigation, as well as a good sentinel species for monitoring studies (Beauvieux et al., 2024; Capó et al., 2022). Juvenile gilthead seabreams were obtained from a local fish farm (Soc. Coop. Acquamarina, Villa Literno, Caserta, Italy) and allocated into three tanks of 1000 L for 7 days. Then, 81 fish were randomized into three groups and acclimatized in their later exposure tanks for 2 weeks before the start of the experiment. Specifically, a total of 27 fish were used for each experimental group. To ensure the credibility of the results, fish of each experimental group were further distributed into 2 different fiberglass 250 L tanks (13 and 14 fish for each tank, respectively). Fish tanks were cleaned daily changing around 10–15 % of the water. The procedure was performed in the morning after feeding to remove any debris, feces, and residual food. The gilthead seabreams were fed twice a day with 1 % (w/w) food per average body weight/tank/day with commercial fish food. The food ration was gradually distributed in each tank to ensure that all fish were able to gain access to the pellets.

2.2. Food preparation and particle characteristics

Commercial food pellets were used to feed fish during the

experiment (VRM S.r.l., NaturAlleva, Verona, Italy). Food pellets had a diameter of 4–4.5 mm, and the ingredient composition is reported in Table S1. In addition to the standard one, two experimental diets were prepared by the same feed company (VRM S.r.l., NaturAlleva, Verona, Italy). The latter had the same ingredient composition as the standard diet with the enrichment of 0.5 or 5 % (w/w) PS-MPs (microparticles GmbH, Berlin, Germany). Specifically, a spherical PS-MP with a diameter between 1 and 20 μm was selected for the experimental procedure. The PS-MP used was a white and odorless powder with a purity of 100 %.

2.3. Fish feeding exposure

The experimental groups were: (1) a control (CON) group fed on the standard diet; (2) a group fed on the experimental diet and exposed to 25 mg/kg b.w./day of PS-MPs (PS1); (3) a group fed on the experimental diet and exposed to 250 mg/kg b.w./day of PS-MPs (PS2). The CON group was fed twice a day with the standard diet, while the PS1 and PS2 groups were fed with the PS-MP enriched diet in the morning, and with the standard diet in the evening. The exposure lasted 21 days. The selected PS-MP doses and the exposure time were chosen based on previous studies conducted by using similar experimental conditions in teleost fish (Ašmonaitė et al., 2018; Jovanović et al., 2018). The latter studies examined the effects of MP particles administered subchronically and chronically through a contaminated diet.

At the end of the experimental time, all fish were euthanized one by one with overexposure to tricaine methane sulfonate (tricaine methanesulfonate-MS222, Sigma Aldrich, St. Louis, MO, USA, 250 ppm). Then, blood was collected through the dorsal artery and all fish were dissected by opening the abdominal cavity. Thus, the posterior intestine, emptied of its contents, and the liver were collected on ice, immediately frozen, and stored at -80°C for subsequent molecular and biochemical analyses. Additionally, a portion of the posterior intestine was also used for gut microbiota-based studies.

2.4. Ethics statement

Animal husbandry and feeding experiments were conducted under license number 1057/2020-PR released by the Italian Ministry of Health and approved by the Ethics Committee of the University of Naples Federico II (Italy). All performed procedures were carried out in the indoor Recirculating Aquaculture System (RAS) at the Department of Veterinary Medicine and Animal Production, University of Naples Federico II, Italy (Italian Ministry of Health authorization n. 25/2019-UT), following the guidelines of the European Communities Council (2010/63/UE). Fish were monitored daily. No mortality was observed, and all fish were apparently healthy during the experiment.

2.5. Biochemical analysis

Blood collected from CON, PS1, and PS2 groups ($n =$ at least 4 each group) was centrifuged at 2500 rpm at 4°C for 12 min, and the obtained serum was stored at -80°C for the following biochemical determinations. Triglycerides (TGL), cholesterol, and total bile acids (TBA) were quantified by using commercial kits (Spinreact, S.A.U., Girona, Spain; Delta Chemie Biotechnology, Naples, Italy).

Liver tissues ($n =$ at least 5 each group) were homogenized in saline solution and then centrifuged at 5000 rpm for 5 min. Supernatants were collected and centrifuged at 14000 rpm at 4°C for 15 min. Subsequently, the obtained supernatants were centrifuged again at 3000 rpm at 4°C for 10 min and then, cholesterol and HDL were quantified by using a commercial kit (Spinreact, S.A.U., Girona, Spain).

2.6. Real-Time semi-quantitative PCR analysis

Total RNA was isolated from the liver ($n =$ at least 7 each group)

using TRIzol Reagent (Bio-Rad Laboratories, Hercules, CA, USA; 7,326,890), and following the extraction kit's protocol for RNA (NucleoSpin®, Macherey-Nagel GmbH & Co, Düren, Germany; FC140955N). Then, cDNA was obtained using a High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Foster City, CA, USA; 4,374,966) from 1 μg total RNA. Real-Time PCRs were performed with a Bio-Rad CFX96 Connect Real-time PCR System instrument and software (Bio-Rad Laboratories, Milan, Italy). The used conditions were previously described (Del Piano et al., 2023b). Each sample contained 500 ng cDNA in $2\times$ QuantiTect SYBR Green PCR Master Mix and primers pairs (IDT Technologies, Coralville, IA, USA) to amplify TLR2 (*TLR2*), TLR5 (*TLR5*), Myeloid differentiation factor 88 (*MyD88*, *MyD88*), Cyclooxygenase (COX)-2 (*COX-2*), Cytochrome (CYP) 27A1 (*CYP27A1*), CYP7B1 (*CYP7B1*), and liver X nuclear receptor (LXR) α (*LXR\alpha*), in a final volume of 50 μL . Data were normalized to *RPS18* as housekeeping genes and analyzed using the $2^{-\Delta\Delta\text{CT}}$ method. Specific primer sequences for target and housekeeping genes are reported in Table 1.

2.7. Gut microbiota-based studies

In this study, the autochthonous microbiota was profiled, comprising the bacteria that colonize the mucosal surface of the digestive tract and make up the core community of the posterior intestine.

This segment was specifically selected because the metabolization of primary BAs into secondary BAs primarily occurs here (Hofmann, 1999). Briefly, the gut specimens were gently squeezed with sterile instruments to remove the luminal contents, in order to avoid the transient free-living bacteria (allochthonous microbiota) associated with the digesta, which were not part of the analysis (Hao and Lee, 2004). The posterior intestine segment of each fish was aseptically opened lengthwise, and the inner walls were scraped to recover mucosal bacteria.

The scraped content of each segment ($n =$ at least 4 selected fish each experimental group) was immediately frozen at -80°C until DNA extraction. The bacterial DNA from each sample was extracted using the QIAamp PowerFecal Pro DNA Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions. High-throughput sequencing of V3–V4 regions of the 16S rDNA gene was performed by Novogene (Cambridge, UK) on an Illumina NovaSeq 6000 platform generating 250 bp paired-end reads. The obtained paired-end reads were demultiplexed and processed using the Quantitative Insights Into Microbial Ecology (QIIME2, version 2021.4 (Bolyen et al., 2019)) software, as previously described (Coretti et al., 2017). Rarefaction curves were used to estimate the completeness of microbial communities sampling; to avoid sample-size biases in subsequent analyses, a sequence rarefaction procedure was applied, setting a depth of 2229 sequences/sample. Statistical significances between alpha and beta diversity indexes were determined using the Kruskal-Wallis test and Analysis of similarities (ANOSIM) with QIIME2 plugins, respectively. The core microbiota for each group was identified by using the QIIME2 script “qiime feature-table core-features”, as the list of genera observed in at least 50 % of the samples within each group. Genera shared between groups were identified in a Venn diagram using Interactive Venn (<http://www.interactivewenn.net/>) (Heberle et al., 2015). The network was generated using the CoNet plugin (Faust and Raes, 2016) for Cytoscape (3.9.1) (Shannon et al., 2003) by employing nonparametric Spearman correlation coefficients with a minimal cut-off threshold of 0.5 ($p < 0.05$, Bonferroni corrected) as the specified parameters. Moreover, to specifically identify Amplicon Sequence Variants (ASVs) associated with marine species, a targeted approach was employed due to the general nature of the SILVA database (SILVA DB) (Quast et al., 2013), which encompasses various environments beyond the marine domain. In this regard, the Basic Local Alignment Search Tool (BLAST) was used in conjunction with the MarRef 1.6 database available on the Marine Metagenomics Portal (<http://mmp2.sfb.uit.no/blast/>).

The phylogenetic investigation of communities by reconstruction of unobserved states (PICRUST2), based on the normalized ASVs table, was

Table 1

Primer sequences used for Real-Time PCR analysis.

Gene name	Gene bank ID	Primer forward (5' → 3')	Primer reverse (5' → 3')
<i>Rps18</i>	AM490061	AGGGTGTGGCAGAGCTTAC	CGCTCAACCTCTCATCAGT
<i>TLR2</i>	Unpublished	TCCATGCTTTCGTCCAGGAC	ACTGTGTTGAGCAAGGCCTC
<i>TLR5</i>	AM296028	CCTGTCTGCAACTGTGAGGA	TGTGGATCTGGTCAAGCTG
<i>Myd88</i>	LR537139.1	GAGGTTGACAGCTGTCTCCC	GGCAGTAGCAGATGAAGGCA
<i>COX-2</i>	AM296029	GAGTACTGGAAGCCGAGCAC	GATATCACTGCCGCTGAGT
<i>CYP27A1</i>	AM885865	TGGCTCTACAAGTTTGGCTTTGA	TGAACCGCAGCGTGTCTTT
<i>CYP7B1</i>	Unpublished	GAGTCGAGTTCAGCAGCGAT	AAGTCCTCTGAGCAACACG
<i>LXRα</i>	FJ502320	GCACTTCGCCTCCAGGACAAG	CAGTCTTCACACAGCCATCAGG

applied to investigate the functional metagenomic profile in each sample (Douglas et al., 2020). The analysis was focused on the Kyoto Encyclopedia of Genes and Genomes (KEGG) ortholog abundances related to the secondary BA production pathway.

2.8. Data and statistical analysis

Data are presented as mean ± SEM. Statistical analysis was performed using analysis of variance (ANOVA) for multiple comparisons followed by Bonferroni's post hoc test, using GraphPad Prism 9 (GraphPad Software, San Diego, CA, USA). Normality was tested using the Shapiro-Wilk test. Statistical significance was set at $p < 0.05$.

3. Results

3.1. PS-MPs perturb the structure of intestinal bacterial community

Using targeted sequencing of the V3-V4 regions of the 16S rRNA gene, we profiled the structure and composition of the intestinal microbiota of PS1 and PS2 fish compared to the CON group. After size filtering, quality control, chimera removal, and exclusion of the ASVs present in only one sample or not phylogenetically recognized, an average number of 12,860 reads/sample (minimum frequency of 2229 reads) was obtained. These reads were clustered into 521 ASVs, representing 19 phyla, 181 genera, and 282 species. The rarefaction curve analysis demonstrated that all samples reached their plateaus, indicating sufficient sequencing depth (data not shown).

The microbial alpha-diversity was measured by using Chao1 (a richness estimator), Shannon index (accounting for both richness and

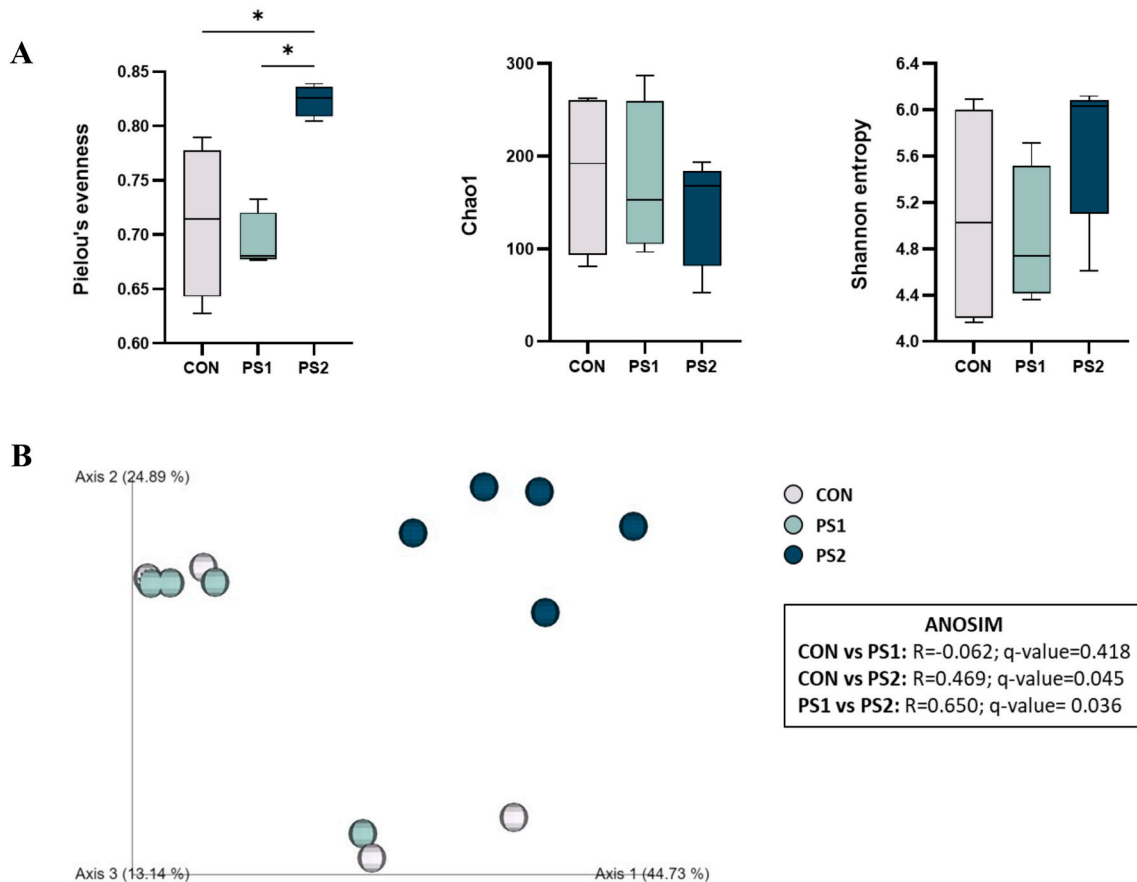


Fig. 1. Impact of PS-MPs on microbial community diversity in *Sparus aurata*. (A) Alpha diversity metrics by means of Pielou's evenness, Chao1 and Shannon entropy ($n =$ at least 4). Each boxplot represents the median, interquartile range, minimum and maximum values. * $q < 0.05$, Kruskal-Wallis pairwise test. (B) Principal Coordinate Analysis (PCoA) plot based on the weighted UniFrac distance matrix. The statistical significance of differences in microbial community composition was assessed using ANOSIM (Analysis of Similarity) with 999 permutations, and results are presented as R statistics and associated p -values ($n =$ at least 4).

evenness), and Pielou's evenness. The CON and PS1 groups showed no significant differences in any of the alpha-diversity indices. However, the PS2 group displayed a significantly increased evenness compared to the CON group, as indicated by the Pielou's evenness index (Fig. 1A). Concerning the beta-diversity analyses, the ANOSIM analysis of the Principal Coordinate Analysis (PCoA) plot built on weighted UniFrac distances revealed that the gut microbiota of the PS2 group deviates from that of all the other groups (Fig. 1B).

3.2. PS-MPs induce alterations in intestinal bacterial community composition

An in-depth analysis was conducted by exploring the core microbiota (genera shared by over 50 % of samples in each group), the distribution of the most abundant genera and species (relative abundance over 0.5 %), and their network of interactions among groups. The core microbiota of the CON, PS1 and PS2 groups consisted of 92, 46 and 51 genera, respectively, showing an impoverishment of core microbiota induced by PS-MPs (Fig. 2A). The group-specific genera among the exposed groups were also studied, which should represent the adaptive modulation of microbiota to PS-MPs: 23 genera were exclusively found in both groups exposed to PS-MPs, with 13/23 specific to the PS2 group core microbiota. In comparison, the CON group presented 53 specific genera. When comparing all groups, a total of 32 overlapping genera were observed, representing the core microbiota able to persist despite changing factors, such as exposure to different doses of MPs (Fig. 2A).

At the genus level, the one-way ANOVA revealed that the observed difference in the microbial assortment between the groups primarily stemmed from significant differences in the abundance of *Prevotella* and *Romboutsia*. Specifically, the PS2 group exhibited lower levels of both genera compared to all other groups. An unidentified genus of the *Peptostreptococcaceae* family was also reduced in the PS2 group compared to the other groups (although the difference was not significant when compared to the CON group), while *Bifidobacterium* was significantly

increased (Fig. 2B).

To ensure a more focused analysis on the desired microbial ecological context, a targeted approach for marine species was employed, with the BLAST function in the MarRef 1.6 database, comprising 1402 records at the time of writing. *Paraclostridium bifermentans* (above reported as *Romboutsia*, based on the SILVA DB) and *Owenweeksia hongkongensis* (SILVA DB: *Prevotella*) were the two most abundant species (accounting for 32.39 % and 25.33 % of the total relative abundance, respectively), while *Clostridium propionicum*, *Clostridium botulinum*, *Labilibaculum antarcticum*, *Echinicola rosea*, and *Micrococcus luteus* presented relative abundances above 0.5 % when considering all groups together. When comparing the relative abundance of these species across all groups using one-way ANOVA, a distinct profile was observed in the PS2 group compared to the other groups: the species *C. propionicum*, *C. botulinum*, *L. antarcticum*, *E. rosea*, and *M. luteus* exhibited elevated levels in comparison to the other groups, while the abundance of *P. bifermentans* and *O. hongkongensis* was decreased (Fig. 2C).

A correlation network was built using the species with a relative abundance >0.5 % to identify patterns of co-occurring bacteria in the complexity of microbial interactions among all groups (Fig. 3A). The network contained 7 nodes (microbial species) connected by 12 significant edges. The two primary species (*P. bifermentans* and *O. hongkongensis*) showed a positive correlation with each other, while the remaining species (*C. propionicum*, *C. botulinum*, *L. antarcticum*, *E. rosea*, and *M. luteus*) formed a module that exerted a negative influence on *P. bifermentans* and *O. hongkongensis*. Fig. 3 also reports the quantitative analysis of betweenness used to identify the species exerting the most influence over the entire network. *P. bifermentans* and *O. hongkongensis* show the highest betweenness centrality, indicating that they are the main contributors to the microbial assortment observed upon PS-MP exposure.

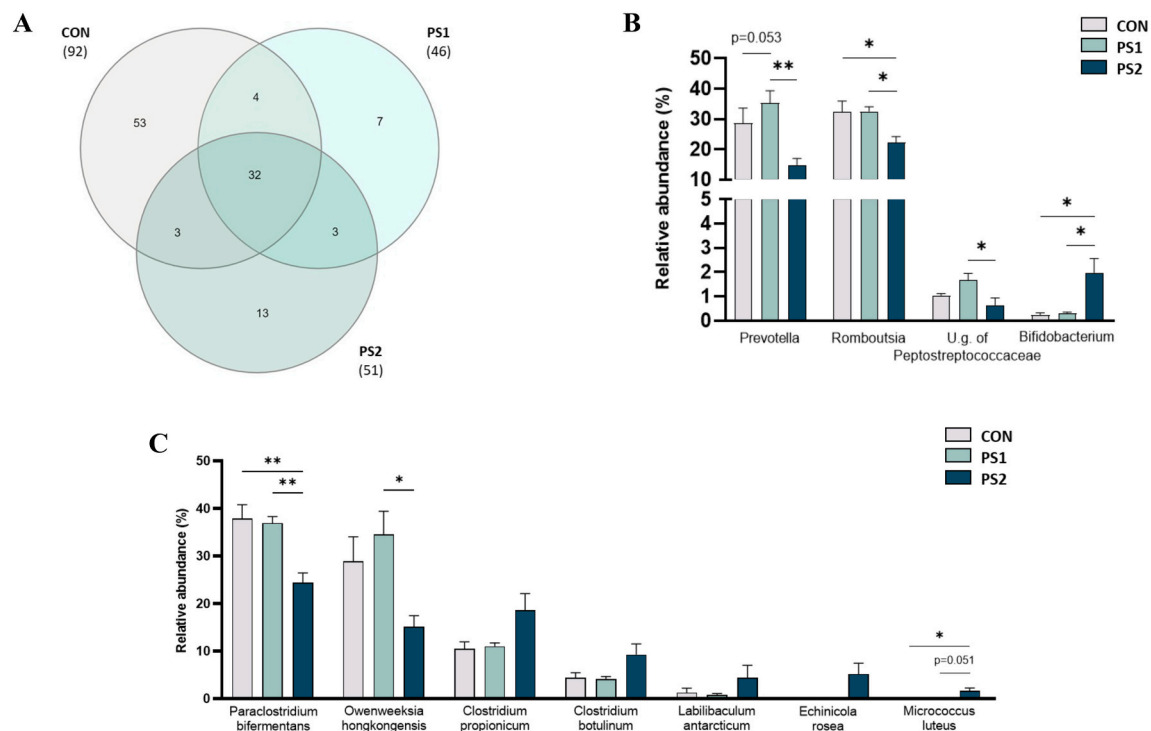


Fig. 2. Effects of PS-MPs on bacterial community composition in *Sparus aurata*. (A) Venn diagram depicting the overlap of bacterial genera within the core microbiotas of each experimental group. (B) Relative abundance of key ASVs discriminating bacterial communities at the genus level, using the SILVA database ($n =$ at least 4). (C) Relative abundance of key ASVs discriminating bacterial communities at the species level, using the MarRef 1.6 database ($n =$ at least 4). Data are shown as mean $\pm$ S.E.M. * $p < 0.05$, ** $p < 0.01$.

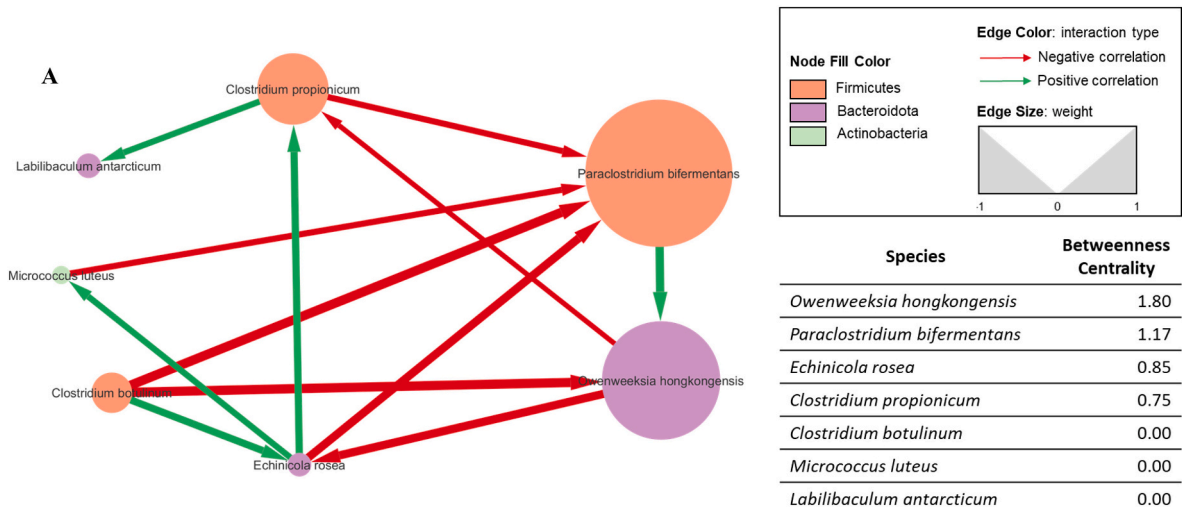


Fig. 3. Co-occurrence network of bacterial species in *Sparus aurata*. (A) Nodes represent individual species and edges represent nonparametric Spearman correlations with a correlation coefficient > 0.5 statistically significant ($p < 0.05$ after Bonferroni correction). Node sizes are proportional to relative abundance, while their colors represent their classification at the phylum level. Species betweenness centrality values are indicated in the associated table.

3.3. PS-MPs modulate the immune response and inflammation in the liver

The hepatic immune response was activated in fish exposed to PS-MPs. Specifically, the TLR levels were impacted by PS-MPs, which increased the mRNA levels of TLR2 at the highest dose (Fig. 4A), and of TLR5 at both doses (Fig. 4B). However, MyD88 gene expression was not modified among the groups (Fig. 4C). On the other hand, COX-2 gene levels were augmented in the PS2 group, indicating an increased inflammatory response after dietary exposure to PS-MPs (Fig. 4D).

3.4. PS-MPs alter serum and hepatic biochemical parameters

PS-MPs increased the levels of circulating TGL. Precisely, serum TGL levels were augmented in gilthead seabreams exposed to the highest dose of PS-MPs (Fig. 5A). On the contrary, serum cholesterol levels were not modified by PS-MPs (Fig. 5B). Moreover, serum TBA were significantly increased in fish of the PS2 group (Fig. 5C). Concomitantly, hepatic cholesterol and HDL showed an increasing trend (Fig. 5D and E), more evident for HDL level at the highest PS-MP dose.

3.5. PS-MPs regulate the BA metabolism in the liver

BA metabolism was impacted by PS-MP exposure in the liver of both PS1 and PS2 groups. In particular, both doses of PS-MPs increased the gene levels of CYP27A1 (Fig. 6A), involved in the primary BA biosynthesis from cholesterol, without modifying those of CYP7B1 (Fig. 6B). Moreover, both doses of PS-MPs increased the mRNA expression of LXR α (Fig. 6C), a receptor that plays a critical role in regulating the expression

of genes involved in hepatic bile and fatty acid synthesis, glucose metabolism, as well as sterol efflux.

3.6. Functional analysis using PICRUSt2 predicts increased secondary BA metabolism upon PS-MP exposure

A PICRUSt2 analysis was subsequently performed to predict the functional capabilities of microbial communities. The KEGGs obtained were filtered to selectively isolate those involved in the secondary BA biosynthesis. The counts of K01442 (Fig. 7A) and K00076 (Fig. 7B), corresponding respectively to cholesterylglucuronide hydrolase (3.5.1.24) and 7 α -hydroxysteroid dehydrogenase (1.1.1.159) enzymes, were significantly higher in PS2 group with respect to the CON group.

4. Discussion

This study describes the effects of PS-MP oral exposure on gut microbiota and liver homeostasis modulated by the gut-liver axis in gilthead seabreams. Exposure to MPs can, directly and indirectly, affect the composition of gut microbiota. Since gut microbiota is shaped by diet components, MP ingestion may alter its community structure in living organisms (Moyo, 2022). Thus, MPs can provide a metabolic base possibly determining the growth of specific microbial taxa. Indirectly, MPs can impact the liver and BA homeostasis, in turn affecting gut microbiota and its functions (Chen et al., 2023; Jin et al., 2019).

Our study showed that MPs induced alterations in gut microbiota composition in gilthead seabream with effects observed mainly at the highest PS-MP dose. Based on previous findings (Xu et al., 2024), higher

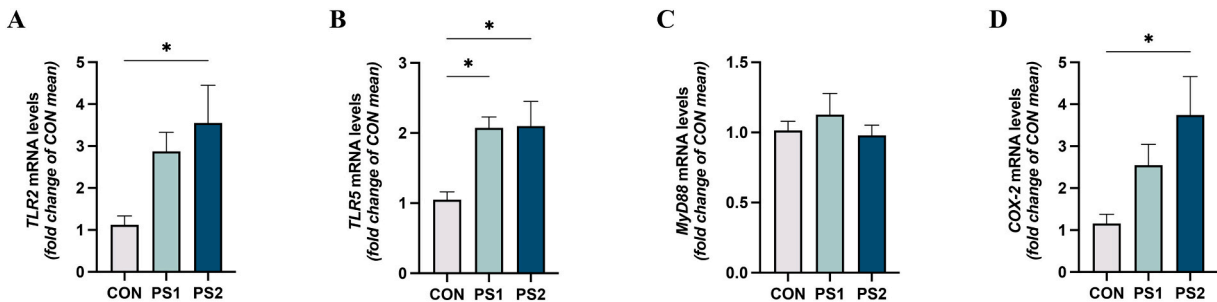


Fig. 4. Impact of PS-MPs on immune and inflammatory-related genes in the liver of *Sparus aurata*. The mRNA expression of (A) TLR2, (B) TLR5, (C) MyD88, and (D) COX-2 in the liver of all experimental groups ($n =$ at least 7). All data are shown as mean $\pm$ S.E.M. * $p < 0.05$.

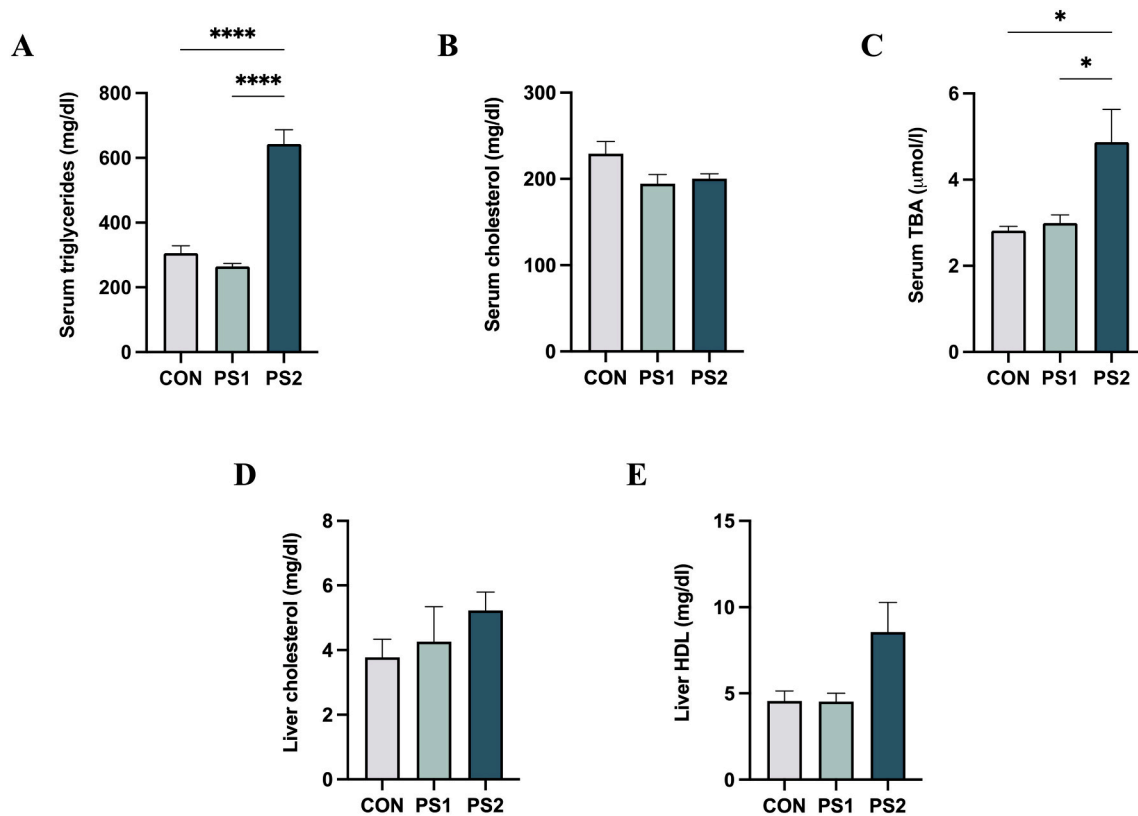


Fig. 5. Effect of PS-MPs on biochemical parameters in serum and liver of *Sparus aurata*. The levels of (A) triglycerides, (B) cholesterol, and (C) TBA in fish serum of all groups (n = at least 4). The levels of (D) cholesterol, and (E) HDL in the liver of gilthead seabreams (n = at least 5). All data are shown as mean $\pm$ S.E.M. *p < 0.05, ****p < 0.0001.

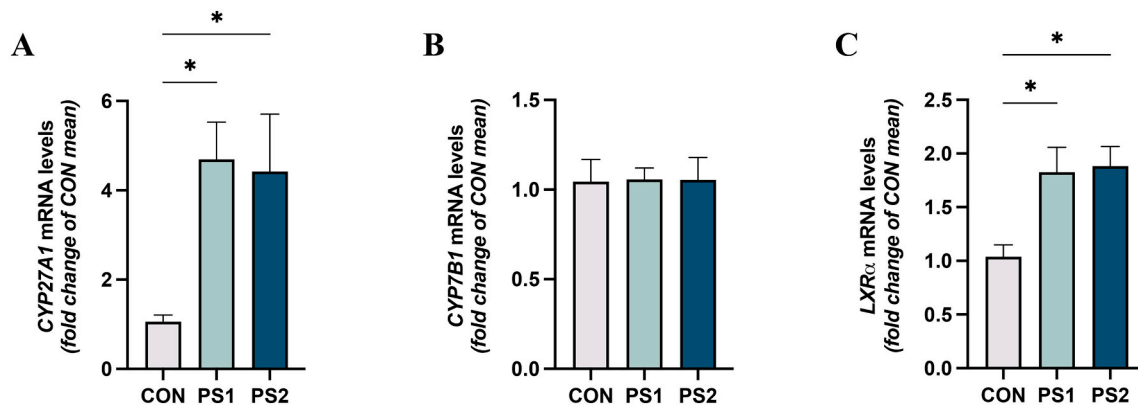


Fig. 6. Modulation of BA metabolism by PS-MPs in the liver of *Sparus aurata*. The mRNA expression of (A) CYP27A1, (B) CYP7B1, and (C) LXR α in the liver of gilthead seabreams (n = 7). All data are shown as mean $\pm$ S.E.M. *p < 0.05.

concentrations of MPs may contribute to prolonged stool residence within the intestinal tract. This, in turn, could create conditions that promote bacterial overgrowth and offer additional colonization platforms, further enhancing gut microbiota remodeling. While the observed evenness enhancement in our PS2 group could be due to a reduction in dominant bacteria, it may also indicate the accidental proliferation of certain species, potentially disrupting the natural balance of the gut microbiota. Similar results were also observed in Asian seabass and mice exposed to high concentrations of MPs (Xie et al., 2021; Lu et al., 2018). When delving into the stability of the gut microbiota of each group, we found that fish receiving PS-MPs exhibited an impoverishment of their core microbiota, suggesting a less conserved configuration of microbial consortia and strong perturbations of

complex microbial assemblages. Our results also demonstrated that, at the species level, *P. bifermentans* (formerly known as *C. bifermentans*) and *O. hongkongensis* relative abundances decreased in the PS2 group, while the species *C. propionicum*, *C. botulinum*, *L. antarcticum*, *E. rosea*, and *M. luteus* exhibited higher levels in comparison to the other groups. It cannot be excluded that the observed reshaping of gut microbiota profiles may represent a sort of adaptation to the PS-MP presence in the intestine being some bacterial species such as *M. luteus* capable of colonizing and utilizing plastics as carbon reservoirs (Gupta et al., 2022; Montazer et al., 2019).

Furthermore, PS-MP exposure reduced the abundance of bacteria commonly found in the gastrointestinal tract of fish such as *P. bifermentans*, reported as positively correlated to butyrate levels in

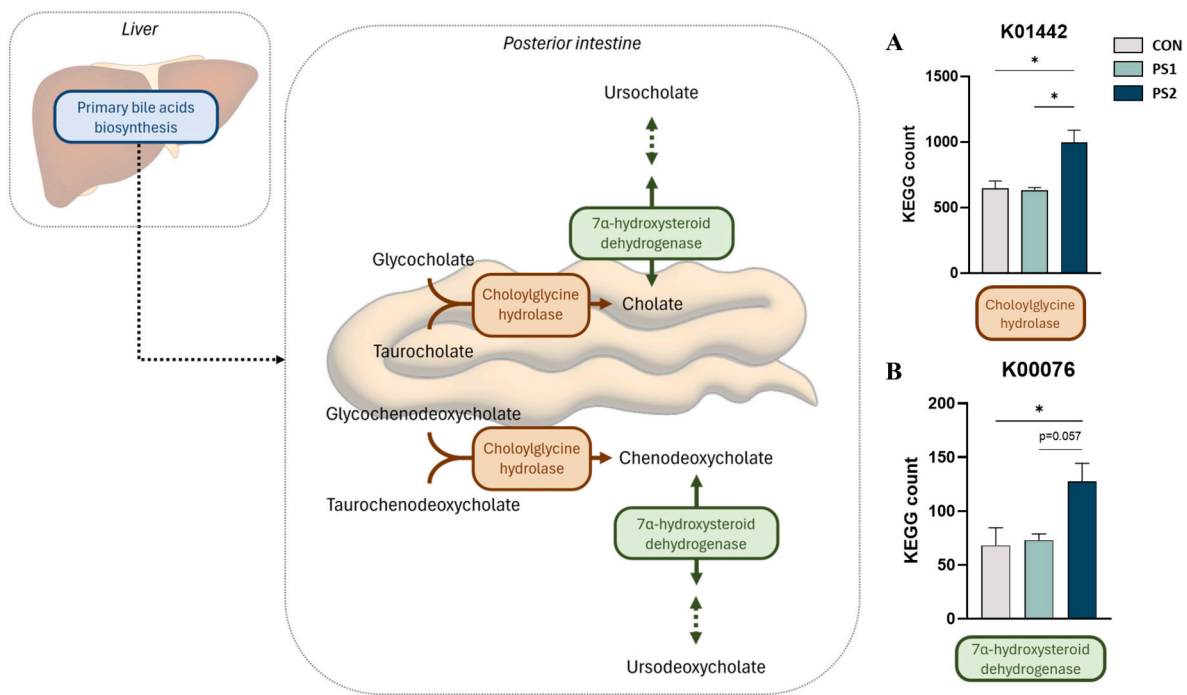


Fig. 7. Impact of PS-MPs on predicted metagenomic functions related to secondary bile acid metabolism. KEGG count of (A) K01442, and (B) K00076 in gilthead seabreams of all experimental groups ($n = \text{at least } 4$). All data are shown as mean $\pm$ S.E.M. * $p < 0.05$.

largemouth bass (Liu et al., 2022). In our network analysis, *P. biferrmentans* was negatively correlated with *M. luteus*, a bacterium naturally found in the aquatic environment, often isolated from the intestinal microflora of various fish species (Akayli et al., 2016; Akayli et al., 2020). *M. luteus* has also been found in diseased fish and related to stress and/or farming conditions (Akayli et al., 2020). In addition, PS2 fish showed an increase of *C. propionicum* and *C. botulinum* levels. *Clostridium*, a common genus within the Firmicutes phylum, is frequently found as a mutualistic symbiont in fish. These bacteria have been shown to contribute to the nutrition of the fish, particularly by supplying fatty acids and vitamins (Clements et al., 2009). However, *Clostridium* is a genus with problematic pathogenic species both for humans and animals, and although no clostridial disease has been described in fish, its presence in the gut is repeatedly reported (Serra et al., 2021).

We previously reported the disruption of intestinal barrier integrity induced by PS-MPs in the posterior intestine of gilthead seabream (Del Piano et al., 2023b). The impairment of intestinal barrier integrity allows the passage of endogenous molecules to the liver through the gut-liver axis. MPs can also directly pass the intestinal barrier and accumulate in the liver (EFSA, 2016; Ma et al., 2021). Recently, we demonstrated the presence of fluorescent PS-MPs (1–20 μm) in the liver of orally exposed gilthead seabream, suggesting that these particles can contribute both directly and indirectly to hepatic damage (Del Piano et al., 2024). The translocation of endogenous molecules and MPs to the liver can activate the TLR pathway, initiating an intracellular signaling cascade and inducing hepatic inflammation (Yang et al., 2022). Indeed, TLRs play a critical role in determining the gut-liver axis immune communication (Kuntzen and Schwabe, 2017). Here, we observed the increased gene expression of both TLR2 and TLR5 at the highest PS-MP dose, and only of TLR5 at the lowest one. Moreover, COX-2 levels were augmented at the highest PS-MP dose, indicating the activation of inflammatory response. Previously, Cui and coauthors observed similar results regarding TLR2 and COX-2 in the carp fish upon MP exposure (Cui et al., 2023).

As known, TLRs are also involved in the onset and progression of hepatic metabolic diseases characterized by increased lipid synthesis and storage (Miura and Ohnishi, 2014). TGL are the main form for

storing and moving fatty acids in the blood and cells. Fatty acid metabolism mainly occurs in the liver. Under normal conditions, the liver stores small amounts of TGL from fatty acids. Environmental pollutants can impair fatty acid metabolism, causing TGL accumulation in the liver and inducing hepatic metabolic diseases (Sun et al., 2022). We previously reported the increased TGL content in the liver of gilthead seabream exposed to PS-MPs (Del Piano et al., 2024). Here, we also showed that serum TGL increased at the highest PS-MP dose, as previously reported in other fish species (Banaei et al., 2022; Zhou et al., 2023). This result suggests the impairment induced by MPs also on systemic TGL homeostasis.

We also reported modifications in the transcription of genes strictly involved in hepatic BA homeostasis. Primary BA biosynthesis occurs in the liver and involves many enzymes and receptors via a classic and an alternative acidic pathway (Chiang, 2004). The acidic pathway aims to produce and maintain the levels of oxysterols. The latter are the oxidized forms of cholesterol that regulate cellular lipid levels and hence cholesterol homeostasis. They are synthesized through the action of key enzymes, including CYP27A1 and CYP7B1, two mitochondrial CYP450 enzymes. CYP27A1 carries out the production of oxysterol 27-hydroxycholesterol, and subsequently, CYP7B1 performs the 7 α -hydroxylation of oxysterols. Hydroxylation reduces the biological activity of oxysterols and enables their conversion to BAs. Therefore, the CYP7B1 enzyme has been recognized as the final regulator of cellular oxysterol concentrations (Pandak and Kakiyama, 2019). To the best of our knowledge, the CYP7B1 expression in gilthead seabream liver was examined here for the first time.

BA synthesis is driven via a positive feedback control mechanism by LXRs, namely LXR α and LXR β . These play crucial functions in the transcriptional regulation of lipid metabolism. High cellular levels of cholesterol and oxysterols induce LXR transcriptional activity, preserving cholesterol homeostasis by controlling the expression of genes involved in cholesterol kinetics and its conversion to BAs (Wang and Tontonoz, 2018).

Here, PS-MPs induced a significant increase in CYP27A1 and LXR α gene expression. These enzymes play a protective role by reducing systemic cholesterol levels through the neosynthesis of primary BAs. We

didn't observe a modulation of liver CYP7B1 mRNA levels. These results suggest that PS-MP exposure induces the transcription of factors that maintain hepatic homeostasis in gilthead seabreams by reducing cholesterol concentration and increasing BA biosynthesis. Indeed, in our conditions, plasma and liver cholesterol were not modified after PS-MP exposure, while serum TBA were increased by the highest PS-MP dose. Similarly, other researchers did not observe changes in cholesterol levels (e.g., Cedervall et al., 2012; Zhao et al., 2020), while others reported an increase (e.g., Banaee et al., 2019; Choi and Kim, 2023; do Nascimento et al., 2023; Hamed et al., 2019) or a decrease (e.g., Haghi and Banaee, 2017; Kim et al., 2021) after MP exposure in fish. However, we observed an increasing trend in liver HDL levels at the highest PS-MP dose, probably due to its potential protective role. Indeed, it is commonly considered “good cholesterol” involved in cholesterol transport and its fecal excretion (Rosenson et al., 2012). MP impact on liver functions and pathways related to BA synthesis can in turn shape gut microbiota and its functions. PICRUST2 mapping of BA-related functions predicted the increase in the PS2 group of K01442 (choleoylglycine hydrolase) involved in the primary BA deconjugation, the first step of secondary BA production, and of K00076 (7 α -hydroxysteroid dehydrogenase), which orientates the secondary BA synthesis towards the production of ursocolate and ursodeoxycholate. Both K01442 and K00076 play an early role in BA transformation, and their PS-MP-induced increase suggests a possible adaptation or co-evolution of gilthead seabream gut microbiota to the modified hepatic BA metabolism. In rodents *Bacteroides*, *Clostridium*, and *Bifidobacterium* are involved in BA deconjugation (Begley et al., 2006). In our study, the disturbance of gilthead seabreams gut microbiota with the increase of *Clostridium* species might have altered the metabolic process of BAs possibly impacting on gut-liver axis.

5. Conclusions

Here we revealed a novel link between the exposure to PS-MPs and the microbiota-gut-liver axis in gilthead seabream used as an experimental model.

Taken together, our results indicated that oral exposure to 1–20 μ m diameter PS-MPs significantly disrupted the gut microbiota architecture in gilthead seabream. This exposure resulted in an impoverishment of the core microbiota, a reduction in predominant bacterial populations, and an increase in low-abundance species, including potentially harmful taxa. Additionally, predicted functions related to BA metabolism were affected. On the other hand, PS-MP exposure stimulated hepatic innate immunity and modulated the expression of hepatic genes involved in BA processing. The study suggests that the well-recognized bidirectional physiological link between the gut and liver contributes to MP-induced toxicity.

These findings could be the basis for hypothesizing beneficial approaches to maintain the proper functioning of the gut-liver axis. In particular, a microbiota-based intervention could be conceived to counteract the potential impact of long-term exposure to MPs due to the ingestion of contaminated food items.

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CRediT authorship contribution statement

Filomena Del Piano: Writing – original draft, Investigation, Formal analysis, Data curation. **Baptiste Mateu:** Writing – original draft, Investigation, Formal analysis, Data curation. **Lorena Coretti:** Writing – original draft, Investigation, Formal analysis. **Luca Borrelli:** Writing – original draft, Methodology, Formal analysis. **Giovanni Piccolo:** Investigation. **Nicola Francesco Addeo:** Investigation. **Sergio Esposito:** Investigation. **Raffaella Mercogliano:** Visualization. **Luigia Turco:** Investigation. **Rosaria Meli:** Supervision. **Francesca Lembo:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Maria Carmela Ferrante:** Writing – review & editing,

Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Ahmed, M.B., Rahman, M.S., Alom, J., Hasan, M.S., Johir, M.A.H., Mondal, M.I.H., Lee, D.Y., Park, J., Zhou, J.L., Yoon, M.H., 2021. Microplastic particles in the aquatic environment: a systematic review. *Sci. Total Environ.* 775, 145793. <https://doi.org/10.1016/j.scitotenv.2021.145793>.
- Akayli, T., Albayrak, G., Ürkü, Ç., Çanak, Ö., Yörük, E., 2016. Characterization of and recovered from common Dentex (*Dentex dentex*) larviculture system. *Mediterr. Mar. Sci.* 17, 163–169.
- Akayli, T., Yardimci, R.E., Ürkü, Ç., 2020. Diagnosis of *Micrococcus luteus* infection in cultured sharpnose sea bream (*Diplodus puntazzo* Cetti, 1777). *Erciyes Üniversitesi Fen Bilimleri Enstitüsü Dergisi* 36, 214–220.
- Aşmonaitė, G., Sundh, H., Asker, N., Carney Almroth, B., 2018. Rainbow trout maintain intestinal transport and barrier functions following exposure to polystyrene microplastics. *Environ. Sci. Technol.* 52 (24), 14392–14401. <https://doi.org/10.1021/acs.est.8b04848>.
- Banaee, M., Soltanian, S., Suredd, A., Gholamhosseini, A., Haghi, B.N., Akhlaghi, M., Derikvandy, A., 2019. Evaluation of single and combined effects of cadmium and micro-plastic particles on biochemical and immunological parameters of common carp (*Cyprinus carpio*). *Chemosphere* 236, 124335. <https://doi.org/10.1016/j.chemosphere.2019.07.066>.
- Banaei, M., Forouzanfar, M., Jafarinia, M., 2022. Toxic effects of polyethylene microplastics on transcriptional changes, biochemical response, and oxidative stress in common carp (*Cyprinus carpio*). *Comp. Biochem. Physiol. C Comp. Pharmacol. Toxicol.* 261, 109423. <https://doi.org/10.1016/j.cbpc.2022.109423>.
- Beauvieux, A., Fromentin, J.M., Romero, D., Couffin, N., Brown, A., Metral, L., Bourjea, J., Bertile, F., Schull, Q., 2024. Molecular fingerprint of gilthead seabream physiology in response to pollutant mixtures in the wild. *Environ. Pollut.* 340 (Pt 2), 122789. <https://doi.org/10.1016/j.envpol.2023.122789>.
- Begley, M., Hill, C., Gahan, C.G.M., 2006. Bile salt hydrolase activity in probiotics. *Appl. Environ. Microbiol.* 72, 1729–1738. <https://doi.org/10.1128/Aem.72.3.1729-1738.2006>.
- Bolyen, E., Rideout, J.R., Dillon, M.R., Bokulich, N.A., Abnet, C.C., Al-Ghathli, G.A., Alexander, H., Alm, E.J., Arumugam, M., Asnicar, F., Bai, Y., Bisanz, J.E., Bittinger, K., Brejnrod, A., Brislawn, C.J., Brown, C.T., Callahan, B.J., Caraballo-Rodriguez, A.M., Chase, J., Cope, E.K., Da Silva, R., Diener, C., Dorrestein, P.C., Douglas, G.M., Durall, D.M., Duvallet, C., Edwards, C.F., Ernst, M., Estaki, M., Fouquier, J., Gauglitz, J.M., Gibbons, S.M., Gibson, D.L., Gonzalez, A., Gorlick, K., Guo, J., Hillmann, B., Holmes, S., Holste, H., Huttenhower, C., Huttley, G.A., Janssen, S., Jarmusch, A.K., Jiang, L., Kaehler, B.D., Kang, K.B., Keefe, C.R., Keim, P., Kelley, S.T., Knights, D., Koester, I., Kosciolk, T., Kreps, J., Langille, M.G.I., Lee, J., Ley, R., Liu, Y.X., Loftfield, E., Lozupone, C., Maher, M., Marotz, C., Martin, B.D., McDonald, D., McIVER, L.J., Melnik, A.V., Metcalf, J.L., Morgan, S.C., Morton, J.T., Naimey, A.T., Navas-Molina, J.A., Nothias, L.F., Orchanian, S.B., Pearson, T.,

- Peoples, S.L., Petras, D., Preuss, M.L., Pruesse, E., Rasmussen, L.B., Rivers, A., Robeson 2nd, M.S., Rosenthal, P., Segata, N., Shaffer, M., Shiffer, A., Sinha, R., Song, S.J., Spear, J.R., Swafford, A.D., Thompson, L.R., Torres, P.J., Trinh, P., Tripathi, A., Turnbaugh, P.J., Ul-Hasan, S., van der Hooft, J.J.J., Vargas, F., Vazquez-Baeza, Y., Vogtmann, E., von Hippel, M., Walters, W., Wan, Y., Wang, M., Warren, J., Weber, K.C., Williamson, C.H.D., Willis, A.D., Xu, Z.Z., Zaneveld, J.R., Zhang, Y., Zhu, Q., Knight, R., Caporaso, J.G., 2019. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat. Biotechnol.* 37, 852–857. <https://doi.org/10.1038/s41587-019-0209-9>.
- Capó, X., Alomar, C., Compa, M., Sole, M., Sanahuja, I., Soliz Rojas, D.L., González, G.P., Garcinuño Martínez, R.M., Deudero, S., 2022. Quantification of differential tissue biomarker responses to microplastic ingestion and plasticizer bioaccumulation in aquaculture reared sea bream *Sparus aurata*. *Environ. Res.* 211, 113063. <https://doi.org/10.1016/j.envres.2022.113063>.
- Cedervall, T., Hansson, L.A., Lard, M., Frohm, B., Linse, S., 2012. Food chain transport of nanoparticles affects behaviour and fat metabolism in fish. *PLoS One* 7, e32254. <https://doi.org/10.1371/journal.pone.0032254>.
- Chen, X., Xu, L., Chen, Q., Su, S., Zhuang, J., Qiao, D., 2023. Polystyrene micro- and nanoparticles exposure induced anxiety-like behaviors, gut microbiota dysbiosis and metabolism disorder in adult mice. *Ecotoxicol. Environ. Saf.* 259, 115000. <https://doi.org/10.1016/j.ecoenv.2023.115000>.
- Chiang, J.Y., 2004. Regulation of bile acid synthesis: pathways, nuclear receptors, and mechanisms. *J. Hepatol.* 40, 539–551. <https://doi.org/10.1016/j.jhep.2003.11.006>.
- Choi, J.H., Kim, J.H., 2023. Toxic effects of sub-acute microplastic (polymide) exposure on the accumulation, hematological, and antioxidant responses in crucian carp, *Carassius carassius*. *Environ. Toxicol. Pharmacol.* 102, 104199. <https://doi.org/10.1016/j.etap.2023.104199>.
- Clements, K.D., Raubenheimer, D., Choat, J.H., 2009. Nutritional ecology of marine herbivorous fishes: ten years on. *Funct. Ecol.* 23, 79–92. <https://doi.org/10.1111/j.1365-2435.2008.01524.x>.
- Cocci, P., Gabrielli, S., Pastore, G., Minicucci, M., Mosconi, G., Palermo, F.A., 2022. Microplastics accumulation in gastrointestinal tracts of *Mullus barbatus* and *Merluccius merluccius* is associated with increased cytokine production and signaling. *Chemosphere* 307, 135813. <https://doi.org/10.1016/j.chemosphere.2022.135813>.
- Coretti, L., Cristiano, C., Florio, E., Scala, G., Lama, A., Keller, S., Cuomo, M., Russo, R., Pero, R., Paciello, O., Mattace Raso, G., Meli, R., Cocozza, S., Calignano, A., Chiariotti, L., Lembo, F., 2017. Sex-related alterations of gut microbiota composition in the BTBR mouse model of autism spectrum disorder. *Sci. Rep.* 7, 45356. <https://doi.org/10.1038/srep45356>.
- Cui, J., Zhang, Y., Liu, L., Zhang, Q., Xu, S., Guo, M.Y., 2023. Polystyrene microplastics induced inflammation with activating the TLR2 signal by excessive accumulation of ROS in hepatopancreas of carp (*Cyprinus carpio*). *Ecotoxicol. Environ. Saf.* 251, 114539. <https://doi.org/10.1016/j.ecoenv.2023.114539>.
- Del Piano, F., Lama, A., Monnolo, A., Pirozzi, C., Piccolo, G., Vozzo, S., De Biase, D., Riccio, L., Fusco, G., Mercogliano, R., Meli, R., Ferrante, M.C., 2023a. Subchronic exposure to polystyrene microplastic differently affects redox balance in the anterior and posterior intestine of *Sparus aurata*. *Animals (Basel)* 13. <https://doi.org/10.3390/ani13040606>.
- Del Piano, F., Lama, A., Piccolo, G., Addeo, N.F., Iaccarino, D., Fusco, G., Riccio, L., De Biase, D., Mattace Raso, G., Meli, R., Ferrante, M.C., 2023b. Impact of polystyrene microplastic exposure on gilthead seabream (*Sparus aurata* Linnaeus, 1758): differential inflammatory and immune response between anterior and posterior intestine. *Sci. Total Environ.* 879, 163201. <https://doi.org/10.1016/j.scitotenv.2023.163201>.
- Del Piano, F., Carney Almroth, B., Lama, A., Piccolo, G., Addeo, N.F., Paciello, O., Martino, G., Esposito, S., Mercogliano, R., Pirozzi, C., Meli, R., Ferrante, M.C., 2024. Subchronic oral exposure to polystyrene microplastics affects hepatic lipid metabolism, inflammation, and oxidative balance in gilthead seabream (*Sparus aurata*). *Ecotoxicol. Environ. Saf.* 279, 116455. <https://doi.org/10.1016/j.ecoenv.2024.116455>.
- do Nascimento, L.S., de Oliveira, S.L., da Costa, C.C., Aracati, M.F., Rodrigues, L.F., Charlie-Silva, I., Conde, G., Mansano, C.F.M., Andreani, D.I.K., de Andrade Belo, M. A., 2023. Deleterious effects of polypropylene microplastic ingestion in Nile tilapia (*Oreochromis niloticus*). *Bull. Environ. Contam. Toxicol.* 111, 13. <https://doi.org/10.1007/s00128-023-03772-y>.
- Douglas, G.M., Maffei, V.J., Zaneveld, J.R., Yurgel, S.N., Brown, J.R., Taylor, C.M., Huttenhower, C., Langille, M.G.I., 2020. PICRUST2 for prediction of metagenome functions. *Nat. Biotechnol.* 38, 685–688. <https://doi.org/10.1038/s41587-020-0548-6>.
- EFSA, 2016. Presence of microplastics and nanoplastics in food, with particular focus on seafood. *EFSA J.* 14, e04501.
- Evariste, L., Barret, M., Mottier, A., Mouchet, F., Gauthier, L., Pinelli, E., 2019. Gut microbiota of aquatic organisms: a key endpoint for ecotoxicological studies. *Environ. Pollut.* 248, 989–999. <https://doi.org/10.1016/j.envpol.2019.02.101>.
- Faust, K., Raes, J., 2016. CoNet app: inference of biological association networks using Cytoscape. *F1000Research* 5. <https://doi.org/10.12688/f1000research.9050.1>.
- Feng, S., Zeng, Y., Cai, Z., Wu, J., Chan, L.L., Zhu, J., Zhou, J., 2021. Polystyrene microplastics alter the intestinal microbiota function and the hepatic metabolism status in marine medaka (*Oryzias latipes*). *Sci. Total Environ.* 759, 143558. <https://doi.org/10.1016/j.scitotenv.2020.143558>.
- Frias, J.P.G.L., Nash, R., 2019. Microplastics: finding a consensus on the definition. *Mar. Pollut. Bull.* 138, 145–147. <https://doi.org/10.1016/j.marpolbul.2018.11.022>.
- Gupta, K.K., Sharma, K.K., Chandra, H., 2022. *Micrococcus luteus* strain CGK112 isolated from cow dung demonstrated efficient biofilm-forming ability and degradation potential toward high-density polyethylene (HDPE). *Arch. Microbiol.* 204, 402. <https://doi.org/10.1007/s00203-022-03023-4>.
- Haghi, B.N., Banaee, M., 2017. Effects of micro-plastic particles on paraquat toxicity to common carp (*Cyprinus carpio*): biochemical changes. *Int. J. Environ. Sci. Technol.* 14, 521–530. <https://doi.org/10.1007/s13762-016-1171-4>.
- Hamed, M., Soliman, H.A.M., Osman, A.G.M., Sayed, A.E.H., 2019. Assessment the effect of exposure to microplastics in Nile Tilapia (*Oreochromis niloticus*) early juvenile: I. Blood biomarkers. *Chemosphere* 228, 345–350. <https://doi.org/10.1016/j.chemosphere.2019.04.153>.
- Hao, W.L., Lee, Y.K., 2004. Microflora of the gastrointestinal tract: a review. *Methods Mol. Biol.* 268, 491–502. <https://doi.org/10.1385/1-59259-766-1:491>.
- Hasegawa, T., Nakaoka, M., 2021. Trophic transfer of microplastics from mysids to fish greatly exceeds direct ingestion from the water column. *Environ. Pollut.* 273, 116468. <https://doi.org/10.1016/j.envpol.2021.116468>.
- Heberle, H., Meirelles, G.V., da Silva, F.R., Telles, G.P., Minghim, R., 2015. InteractiVenn: a web-based tool for the analysis of sets through Venn diagrams. *BMC Bioinforma.* 16, 1–7. <https://doi.org/10.1186/s12859-015-0611-3>.
- Hofmann, A.F., 1999. The continuing importance of bile acids in liver and intestinal disease. *Arch. Intern. Med.* 159 (22), 2647–2658. <https://doi.org/10.1001/archinte.159.22.2647>.
- Hofmann, A.F., Hagey, L.R., Krasowski, M.D., 2010. Bile salts of vertebrates: structural variation and possible evolutionary significance. *J. Lipid Res.* 51 (2), 226–246. <https://doi.org/10.1194/jlr.R000042>.
- Hu, J., Zuo, J., Li, J., Zhang, Y., Ai, X., Zhang, J., Gong, D., Sun, D., 2022. Effects of secondary polyethylene microplastic exposure on crucian (*Carassius carassius*) growth, liver damage, and gut microbiome composition. *Sci. Total Environ.* 802, 149736. <https://doi.org/10.1016/j.scitotenv.2021.149736>.
- Huang, Z., Weng, Y., Shen, Q., Zhao, Y., Jin, Y., 2021. Microplastic: a potential threat to human and animal health by interfering with the intestinal barrier function and changing the intestinal microenvironment. *Sci. Total Environ.* 785, 147365. <https://doi.org/10.1016/j.scitotenv.2021.147365>.
- Issac, M.N., Kandasubramanian, B., 2021. Effect of microplastics in water and aquatic systems. *Environ. Sci. Pollut. Res. Int.* 28 (16), 19544–19562. <https://doi.org/10.1007/s11356-021-13184-2>.
- Jin, Y., Lu, L., Tu, W., Luo, T., Fu, Z., 2019. Impacts of polystyrene microplastic on the gut barrier, microbiota and metabolism of mice. *Sci. Total Environ.* 649, 308–317. <https://doi.org/10.1016/j.scitotenv.2018.08.353>.
- Jovanović, B., Gökdag, K., Güven, O., Emre, Y., Whitley, E.M., Kideys, A.E., 2018. Virgin microplastics are not causing imminent harm to fish after dietary exposure. *Mar. Pollut. Bull.* 130, 123–131. <https://doi.org/10.1016/j.marpolbul.2018.03.016>.
- Kim, J.H., Yu, Y.B., Choi, J.H., 2021. Toxic effects on bioaccumulation, hematological parameters, oxidative stress, immune responses and neurotoxicity in fish exposed to microplastics: a review. *J. Hazard. Mater.* 413, 125423. <https://doi.org/10.1016/j.jhazmat.2021.125423>.
- Kuntzen, C., Schwabe, R.F., 2017. Gut microbiota and Toll-like receptors set the stage for cytokine-mediated failure of antibacterial responses in the fibrotic liver. *Gut* 66, 396–398. <https://doi.org/10.1136/gutjnl-2016-312486>.
- Le, L.T., Nguyen, K.N., Nguyen, P.T., Duong, H.C., Bui, X.T., Hoang, N.B., Nghiem, L.D., 2022. Microfibers in laundry wastewater: problem and solution. *Sci. Total Environ.* 852, 158412. <https://doi.org/10.1016/j.scitotenv.2022.158412>.
- Li, A., Wang, Y., Kulyar, M.F., Iqbal, M., Lai, R., Zhu, H., Li, K., 2023. Environmental microplastics exposure decreases antioxidant ability, perturbs gut microbial homeostasis and metabolism in chicken. *Sci. Total Environ.* 856, 159089. <https://doi.org/10.1016/j.scitotenv.2022.159089>.
- Li, B., Ding, Y., Cheng, X., Sheng, D., Xu, Z., Rong, Q., Wu, Y., Zhao, H., Ji, X., Zhang, Y., 2020. Polyethylene microplastics affect the distribution of gut microbiota and inflammation development in mice. *Chemosphere* 244, 125492. <https://doi.org/10.1016/j.chemosphere.2019.125492>.
- Liu, Y., Cao, Y., Zhang, Y., Fan, J., Zhou, H., Huang, H., Jiang, W., Zhang, W., Deng, J., Tan, B., 2022. Intestinal flora and immunity response to different viscous diets in juvenile largemouth bass, *Micropterus salmoides*. *Fish Shellfish Immunol.* 127, 1012–1023. <https://doi.org/10.1016/j.fsi.2022.06.054>.
- Lu, L., Wan, Z., Luo, T., Fu, Z., Jin, Y., 2018. Polystyrene microplastics induce gut microbiota dysbiosis and hepatic lipid metabolism disorder in mice. *Sci. Total Environ.* 631–632, 449–458. <https://doi.org/10.1016/j.scitotenv.2018.03.051>.
- Ma, C., Chen, Q., Li, J., Li, B., Liang, W., Su, L., Shi, H., 2021. Distribution and translocation of micro- and nanoplastics in fish. *Crit. Rev. Toxicol.* 51, 740–753. <https://doi.org/10.1080/10408444.2021.2024495>.
- McIlwraith, H.K., Kim, J., Helm, P., Bhavsar, S.P., Metzger, J.S., Rochman, C.M., 2021. Evidence of microplastic translocation in wild-caught fish and implications for microplastic accumulation dynamics in food webs. *Environ. Sci. Technol.* 55, 12372–12382. <https://doi.org/10.1021/acs.est.1c02922>.
- Miura, K., Ohnishi, H., 2014. Role of gut microbiota and Toll-like receptors in nonalcoholic fatty liver disease. *World J. Gastroenterol.* 20, 7381–7391. <https://doi.org/10.3748/wjg.v20.i23.7381>.
- Montazer, Z., Habibi Najafi, M.B., Levin, D.B., 2019. Microbial degradation of low-density polyethylene and synthesis of polyhydroxyalkanoate polymers. *Can. J. Microbiol.* 65, 224–234. <https://doi.org/10.1139/cjm-2018-0335>.
- Moyo, S., 2022. An enigma: a meta-analysis reveals the effect of ubiquitous microplastics on different taxa in aquatic systems. *Front. Environ. Sci.* 10, 999349. <https://doi.org/10.3389/fenvs.2022.999349>.
- Pandak, W.M., Kakiyama, G., 2019. The acidic pathway of bile acid synthesis: not just an alternative pathway (*). *Liver Res.* 3, 88–98. <https://doi.org/10.1016/j.livres.2019.05.001>.
- Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., Glockner, F.O., 2013. The SILVA ribosomal RNA gene database project: improved

- data processing and web-based tools. *Nucleic Acids Res.* 41, D590–D596. <https://doi.org/10.1093/nar/gks1219>.
- Rosenon, R.S., Brewer, H.B., Davidson, W.S., Fayad, Z.A., Fuster, V., Goldstein, J., Hellerstein, M., Jiang, X.C., Phillips, M.C., Rader, D.J., Remaley, A.T., Rothblat, G. H., Tall, A.R., Yvan-Charvet, L., 2012. Cholesterol efflux and atheroprotection advancing the concept of reverse cholesterol transport. *Circulation* 125, 1905–1919. <https://doi.org/10.1161/Circulationaha.111.066589>.
- Ruiz, A., Andree, K.B., Furones, D., Holhorea, P.G., Caldach-Giner, J.À., Viñas, M., Pérez-Sánchez, J., Gisbert, E., 2023. Modulation of gut microbiota and intestinal immune response in gilthead seabream (*Sparus aurata*) by dietary bile salt supplementation. *Front. Microbiol.* 14, 1123716. <https://doi.org/10.3389/fmicb.2023.1123716>.
- Santos, A.L., Rodrigues, C.C., Oliveira, M., Rocha, T.L., 2022. Microbiome: a forgotten target of environmental micro(nano)plastics? *Sci. Total Environ.* 822, 153628. <https://doi.org/10.1016/j.scitotenv.2022.153628>.
- Serra, C.R., Oliva-Teles, A., Enes, P., Tavares, F., 2021. Gut microbiota dynamics in carnivorous European seabass (*Dicentrarchus labrax*) fed plant-based diets. *Sci. Rep.* 11, 447. <https://doi.org/10.1038/s41598-020-80138-y>.
- Shannon, P., Markiel, A., Ozier, O., Baliga, N.S., Wang, J.T., Ramage, D., Amin, N., Schwikowski, B., Ideker, T., 2003. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Res.* 13, 2498–2504. <https://doi.org/10.1101/gr.1239303>.
- Sun, J., Fang, R., Wang, H., Xu, D.X., Yang, J., Huang, X., Cozzolino, D., Fang, M., Huang, Y., 2022. A review of environmental metabolism disrupting chemicals and effect biomarkers associating disease risks: where exposomics meets metabolomics. *Environ. Int.* 158, 106941. <https://doi.org/10.1016/j.envint.2021.106941>.
- Sun, Y., Deng, Q., Zhang, Q., Zhou, X., Chen, R., Li, S., Wu, Q., Chen, H., 2024. Hazards of microplastics exposure to liver function in fishes: a systematic review and meta-analysis. *Mar. Environ. Res.* 196, 106423. <https://doi.org/10.1016/j.marenvres.2024.106423>.
- Tilg, H., Adolph, T.E., Trauner, M., 2022. Gut-liver axis: pathophysiological concepts and clinical implications. *Cell Metab.* 34, 1700–1718. <https://doi.org/10.1016/j.cmet.2022.09.017>.
- Wang, B., Tontonoz, P., 2018. Liver X receptors in lipid signalling and membrane homeostasis. *Nat. Rev. Endocrinol.* 14, 452–463. <https://doi.org/10.1038/s41574-018-0037-x>.
- Wen, J., Sun, H., Yang, B., Song, E., Song, Y., Jiang, G., 2024. Environmentally relevant concentrations of microplastic exposure cause cholestasis and bile acid metabolism dysregulation through a gut-liver loop in mice. *Environ. Sci. Technol.* 58, 1832–1841. <https://doi.org/10.1021/acs.est.3c07108>.
- Xie, M.J., Lin, L., Xu, P., Zhou, W.G., Zhao, C.S., Ding, D.W., Suo, A.N., 2021. Effects of microplastic fibers on *Lates calcarifer* juveniles: accumulation, oxidative stress, intestine microbiome dysbiosis and histological damage. *Ecol. Indic.* 133, 108370. <https://doi.org/10.1016/j.ecolind.2021.108370>.
- Xu, R., Cao, J.W., Lv, H.L., Geng, Y., Guo, M.Y., 2024. Polyethylene microplastics induced gut microbiota dysbiosis leading to liver injury via the TLR2/NF- κ B/NLRP3 pathway in mice. *Sci. Total Environ.* 917, 170518. <https://doi.org/10.1016/j.scitotenv.2024.170518>.
- Xu, S., Ma, J., Ji, R., Pan, K., Miao, A.J., 2020. Microplastics in aquatic environments: occurrence, accumulation, and biological effects. *Sci. Total Environ.* 703, 134699. <https://doi.org/10.1016/j.scitotenv.2019.134699>.
- Yang, W., Jannatun, N., Zeng, Y., Liu, T., Zhang, G., Chen, C., Li, Y., 2022. Impacts of microplastics on immunity. *Front. Toxicol.* 4, 956885. <https://doi.org/10.3389/ftox.2022.956885>.
- Yin, J., Ju, Y., Qian, H., Wang, J., Miao, X., Zhu, Y., Zhou, L., Ye, L., 2022. Nanoplastics and microplastics may be damaging our livers. *Toxics* 10. <https://doi.org/10.3390/toxics10100586>.
- Yin, K., Wang, D., Zhang, Y., Lu, H., Wang, Y., Xing, M., 2023. Dose-effect of polystyrene microplastics on digestive toxicity in chickens (*Gallus gallus*): multi-omics reveals critical role of gut-liver axis. *J. Adv. Res.* 52, 3–18. <https://doi.org/10.1016/j.jare.2022.10.015>.
- Zhao, P., Lu, W., Avellan-Llaguno, R.D., Liao, X., Ye, G., Pan, Z., Hu, A., Huang, Q., 2023. Gut microbiota related response of *Oryzias melastigma* to combined exposure of polystyrene microplastics and tetracycline. *Sci. Total Environ.* 905, 167359. <https://doi.org/10.1016/j.scitotenv.2023.167359>.
- Zhao, Y., Bao, Z., Wan, Z., Fu, Z., Jin, Y., 2020. Polystyrene microplastic exposure disturbs hepatic glycolipid metabolism at the physiological, biochemical, and transcriptomic levels in adult zebrafish. *Sci. Total Environ.* 710, 136279. <https://doi.org/10.1016/j.scitotenv.2019.136279>.
- Zhou, W., Shi, W., Du, X., Han, Y., Tang, Y., Ri, S., Ju, K., Kim, T., Huang, L., Zhang, W., Yu, Y., Tian, D., Yu, Y., Chen, L., Wu, Z., Liu, G., 2023. Assessment of nonalcoholic fatty liver disease symptoms and gut-liver axis status in zebrafish after exposure to polystyrene microplastics and oxytetracycline, alone and in combination. *Environ. Health Perspect.* 131, 47006. <https://doi.org/10.1289/EHP11600>.